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SUPPLEMENTING SENSOR DATA FOR PROCESSING USING AI SYSTEMS AND APPLICATIONS

Abstract

In various examples, processing sensor data using rankings for AI systems and applications is described herein. Systems and methods are disclosed that determine rankings for sensor data, where the rankings may then be used to process the sensor data. For instance, a device that generates sensor data using a sensor may determine rankings for various portions of the sensor data, such as by analyzing the sensor data and/or related sensor data to detect configured events. For example, if the sensor data includes image data, then the device may determine a respective ranking for different groups of frames that are associated with different events. A system(s) that then use the rankings when processing the sensor data using one or more processing tasks. For example, the system(s) may determine which processing tasks to use for processing different portions of the sensor data based at least on the rankings.

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Background/Summary

BACKGROUND

[0001] Systems use processing pipelines to process sensor data received from devices, where the processing pipelines are configured to perform various processing tasks on the sensor data in order to generate outputs. For example, a system may use an image processing pipeline to process image data generated using a camera, where the image processing pipeline performs different types of image processing tasks such as inferencing, object detection, object tracking, and/or the like. Based at least on the configurations of the devices and/or the systems, these processing pipelines uniformly process an entirety of the sensor data (e.g., every frame of image data) to perform the processing tasks.

[0002] However, in some circumstances, not every portion of sensor data may be necessary to perform one or more of processing tasks in a processing pipeline. For example, if the image data in the example above represents a number of frames that do not depict objects and/or motion, then the image processing pipeline may not need to perform object detection or object tracking, and thus can skip (or reduce) the processing of at least some of these frames. As such, in some circumstances, these systems may utilize a greater amount of computing resources (e.g., computer processing unit resources, graphics processing unit resources, memory resources, etc.) for processing sensor data than is required to generate a desired output. Additionally, this over utilization of computing resources may increase when systems are processing sensor data in batches from multiple sensors and/or multiple devices.

SUMMARY

[0003] Embodiments of the present disclosure relate to supplementing sensor data processing for processing in AI systems and applications. Systems and methods are disclosed that determine significance indicia (e.g., rankings) for various portions of sensor data, where the rankings may then be used to process the sensor data (e.g., using a processing pipeline). For instance, a device that generates sensor data using a sensor may determine a hierarchy for various portions of the sensor data, such as by analyzing the sensor data and/or related sensor data to detect configured events. For example, if the sensor data includes image data, then the device may determine a respective hierarchical ranking for different groups of frames that are associated with different events. A system(s) that then use the hierarchical rankings when processing the sensor data using one or more processing tasks. For example, the system(s) may determine which processing tasks to use for processing different portions of the sensor data based at least on the hierarchy.

[0004] In contrast to conventional systems, such as those described above, the current systems, in some embodiments, determine one or more hierarchies for various portions of sensor data, such as groups of frames of the sensor data, where the hierarchical rankings are then used during processing to select different processing tasks for the portions of the sensor data. This way, the current systems may not need to uniformly process all of the sensor data using the same processing tasks, which may save computing resources and/or increase the throughput of the processing by refraining from unnecessarily processing some portions of the sensor data using specific processing tasks. For example, instead of performing inferencing, object detection, and/or object tracking for every frame of image data, the current systems may refrain from processing at least a portion of the frames that do not depict objects and/or motion using these processing tasks. As will be described in more detail herein, these improvements of saving computing resources and/or increasing throughput may increase based at least on the number of devices and/or the amount of sensor data for which processing is being performed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The present systems and methods for supplementing sensor data for processing in AI systems and applications are described in detail below with reference to the attached drawing figures, wherein:

[0006] FIG. **1** illustrates an example of a process for determining rankings for sensor data, where the rankings are then used to process the sensor data, in accordance with some embodiments of the present disclosure;

[0007] FIG. **2** illustrates an example of configuration data that may be used for ranking sensor data, in accordance with some embodiments of the present disclosure;

[0008] FIG. **3** illustrates an example of processing image data to detect events, in accordance with some embodiments of the present disclosure;

[0009] FIG. **4** illustrates an example of ranking images based at least on detected events, in accordance with some embodiments of the present disclosure;

[0010] FIG. **5** illustrates an example of processing frames using different processing components, where the ranking is based at least on rankings associated with the frames, in accordance with some embodiments of the present disclosure;

[0011] FIG. **6** is a flow diagram showing a method for ranking sensor data, in accordance with some embodiments of the present disclosure;

[0012] FIG. **7** is a flow diagram showing a method for using one or more rankings associated with sensor data in order to process the sensor data, in accordance with some embodiments of the present disclosure;

[0013] FIG. **8** is a block diagram of an example content streaming system suitable for use in implementing some embodiments of the present disclosure;

[0014] FIG. **9** is a block diagram of an example computing device suitable for use in implementing some embodiments of the present disclosure; and

[0015] FIG. **10** is a block diagram of an example data center suitable for use in implementing some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0016] Systems and methods are disclosed related to supplementing sensor data for processing in AI systems and applications. For instance, a device may generate sensor data using one or more sensors. As described herein, the sensor data may include, but is not limited to, image data generated using one or more image sensors (e.g., one or more cameras), audio data generated using one or more microphones, LiDAR data generated using one or more LiDAR sensors, RADAR data generated using one or more RADAR sensors, motion data generated using one or more motion sensors (e.g., one or more passive infrared sensors, etc.), and/or any other type of sensor data generated using any other type of sensor. The device may then send at least a portion of the sensor data to one or more systems for processing. For example, the device may send the image data to the system(s) that is configured to process the image data using one or more image processing pipelines.

[0017] In some examples, the device may rank at least a portion of the sensor data before sending the sensor data to one or more computing device(s) (e.g., the system(s)). For instance, the device may determine a respective ranking for different groups of sensor representations (e.g., groups of frames), where a group of sensor representations may include one sensor representation, five sensor representations, ten sensor representations, thirty sensor representations, and/or any other number of sensor representations. For a first example, when determining rankings for frames of image data, the device may determine a respective ranking for each frame represented by the image data. For a second example, and again when determining rankings for frames of image data, the device may determine a respective ranking for each group of frames represented by the image data, such as based on a ranking associated with one of the frames (e.g., the first frame) included in the group of frames.

[0018] To determine the rankings, the device may receive, obtain, generate, and/or be configured with data (referred to, in some examples, as “configuration data”) representing information for ranking the sensor representations. For instance, the configuration data may represent at least a first ranking associated with a first event, a second ranking associated with a second event, a third ranking associated with a third event, and/or so forth. For example, when ranking frames, the configuration data may indicate or include rankings associated with slow motion being detected, fast motion being detected, an object being detected, low light being detected, custom weightage, no motion being detected, and/or any other ranking types associated with other event types. The device may then use this configuration data and/or one or more processing techniques to rank sensor data.

[0019] For instance, in some examples, the device may process at least a portion of the sensor data for which the device is ranking to detect one or more of the events associated with the configuration data. In some examples, when processing the sensor data, the device may use one or more machine learning models, one or more neural networks, one or more algorithms, one or more modules, one or more software applications, one or more hardware components, and/or any other type of processing component that is configured to process sensor data to perform a task. Additionally, as will be described in more detail herein, the processing component(s) used by the device may require fewer computing resources as compared to one or more processing components later used by the system(s) that then uses the rankings to process the sensor data. For example, if the device is processing the sensor data using one or more machine learning models associated with object and/or motion detection, then the machine learning model(s) may include fewer layers (e.g., include a “lightweight” machine learning model) as compared to one or more machine learning models later used by the system(s) to perform object and/or motion detection.

[0020] For an example of processing sensor data, and if the sensor data again includes image data, the device may process at least a portion of the image data to determine whether a frame is associated with one or more of the events represented by the configuration data. For instance, the device may determine whether the frame depicts an object, depicts low light, is associated with motion, and/or is associated with any other type of event. Based at least on determining that the frame is associated with an event, such as a detected object, the device may then rank the frame using the ranking that is associated with the event, such as the third ranking in the example above. In some examples, the device may perform similar processes to rank each of the other frames of the image data. Additionally, or alternatively, in some examples, the device may then use that ranking for a group of frames, such as a number of next frames (e.g., one next frame, five next frames, ten next frames, thirty next frames, etc.) of the image data. The device may then continue to perform similar processes to rank additional groups of frames of the image data.

[0021] Additionally to, or alternatively from, processing the sensor data for which the device is ranking, in some examples, the device may process additional sensor data to detect one or more of the events associated with the configuration data. For example, and again if the sensor data being ranked includes image data, the device may process motion data to detect an event, such as a presence of an object being detected or an object not being detected (e.g., an absence of an object). The device may then associate at least a portion of the image data with the detected event. For example, the device may associate one or more frames of the image data, which were generated at an approximately same time as the motion data, with the event. The device may then perform one or more of the processes described herein to rank the frame(s) of the image data based at least on the associated event. Additionally, in some examples, the device may perform similar processes for other types of additional sensor data that may be used to detect one or more of the events, such as audio data, LiDAR data, RADAR data, and/or the like.

[0022] The device may then generate data (referred to, in some examples, as “ranking data”) representing the rankings for the sensor data and send the ranking data along with the sensor data to the system(s) for processing. As described herein, the device may use any technique to send the ranking data along with the sensor data to the system(s). For example, and again if the sensor data includes image data, the device may embed the rankings into messages (e.g., supplemental enhanced information messages, etc.) associated with the image data and/or embed the rankings using dedicates fields of the image data. The device may also encode the image data, either before and/or after the embedding, to the generated encoded image data. After performing this processing, the device may send the encoded image data along with the embedded rankings to the system(s) for processing.
[0023] The system(s) may then use the rankings when processing the sensor data using one or more processing tasks, where the processing task(s) may be associated with a processing pipeline. For instance, the system(s) may determine which processing task(s) to use when processing portions of the sensor data, which processing task(s) to refrain from using when processing portions of the sensor data, additional processing task(s) (e.g., outside of the processing pipeline) to use to process the portions of the sensor data, and/or the like using the rankings. For a first example, and again if the sensor data includes image data, if a first ranking associated with one or more first frames indicates that the first frame(s) is associated with an object being detected, then the system(s) may process the first frame(s) using a processing task(s) associated with inferencing, object detection, and/or object tracking. However, if a second ranking associated with one or more second frames indicates that the second frame(s) is not associated with an object being detected, then the system(s) may refrain from processing the second frame(s) using the same processing tasks(s).
[0024] For a second example, and again if the sensor data includes image data, if a first ranking associated with one or more first frames indicates that the first frame(s) is associated with motion from a single object, then the system(s) may process the first frame(s) using a first processing task(s) associated with inferencing, object detection, and/or object tracking. However, if a second ranking associated with one or more second frames indicates that the second frame(s) is associated with motion from multiple objects, then the system(s) may process the second frame(s) using a second processing tasks(s) that is also associated with inferencing, object detection, and/or object tracking. However, in this second example, the second processing task(s) may include higher quality machine learning models, neural networks, algorithms, and/or the like as compared to the first processing task(s) such that the second processing task(s) is better able to process sensor data that is associated with multiple objects.
[0025] The systems and methods described herein may be used by, without limitation, non-autonomous vehicles or machines, semi-autonomous vehicles or machines (e.g., in one or more adaptive driver assistance systems (ADAS)), autonomous vehicles or machines, piloted and un-piloted robots or robotic platforms, warehouse vehicles, off-road vehicles, vehicles coupled to one or more trailers, flying vessels, boats, shuttles, emergency response vehicles, motorcycles, electric or motorized bicycles, aircraft, construction vehicles, underwater craft, drones, and/or other vehicle types. Further, the systems and methods described herein may be used for a variety of purposes, by way of example and without limitation, for machine control, machine locomotion, machine driving, synthetic data generation, model training, perception, augmented reality, virtual reality, mixed reality, robotics, security and surveillance, simulation and digital twinning, autonomous or semi-autonomous machine applications, deep learning, environment simulation, object or actor simulation and/or digital twinning, data center processing, conversational AI, light transport simulation (e.g., ray-tracing, path tracing, etc.), collaborative content creation for 3D assets, cloud computing and/or any other suitable applications.
[0026] Disclosed embodiments may be comprised in a variety of different systems such as automotive systems (e.g., a control system for an autonomous or semi-autonomous machine, a perception system for an autonomous or semi-autonomous machine), systems implemented using a

robot, aerial systems, medial systems, boating systems, smart area monitoring systems, systems for performing deep learning operations, systems for performing simulation operations, systems for performing digital twin operations, systems implemented using an edge device, systems implementing large language models (LLMs), systems incorporating one or more virtual machines (VMs), systems for performing synthetic data generation operations, systems implemented at least partially in a data center, systems for performing conversational AI operations, systems for performing light transport simulation, systems for performing collaborative content creation for 3D assets, systems for performing generative AI operations, systems implemented at least partially using cloud computing resources, and/or other types of systems.

[0027] With reference to FIG. **1**, FIG. **1** illustrates an example of a process **100** for determining rankings for sensor data, where the rankings are then used to process the sensor data, in accordance with some embodiments of the present disclosure. It should be understood that this and other arrangements described herein are set forth only as examples. Other arrangements and elements (e.g., machines, interfaces, functions, orders, groupings of functions, etc.) may be used in addition to or instead of those shown, and some elements may be omitted altogether. Further, many of the elements described herein are functional entities that may be implemented as discrete or distributed components or in conjunction with other components, and in any suitable combination and location. Various functions described herein as being performed by entities may be carried out by hardware, firmware, and/or software. For instance, various functions may be carried out by a processor executing instructions stored in memory.

[0028] The process **100** may include at least one device **102** generating sensor data **104** using one or more sensors **106**. As described herein, the sensor data **104** may include, but is not limited to, image data generated using one or more image sensors (e.g., one or more cameras), audio data generated using one or more microphones, LiDAR data generated using one or more LiDAR sensors, RADAR data generated using one or more RADAR sensors, motion data generated using one or more motion sensors (e.g., one or more passive infrared sensors, etc.), and/or any other type of sensor data generated using any other type of sensor. In some examples, the sensor data **104** may represent sensor representations. For example, and for image data, the image data may represent frames. Additionally, in some examples, the sensor data **104** may represent other types of information, such as timestamps indicating when portions of the sensor data **104** (e.g., sensor representations) are generated using the sensor(s) **106**. For example, and again for the image data, the image data may represent a respective timestamp when one or more frames of the image data are generated using the image sensor(s).

[0029] As described herein, the device **102** may then perform one or more techniques in order to rank one or more portions of the sensor data **104**. For instance, the device **102** may use configuration data **108** representing information for ranking the sensor representations of the sensor data **104**. In some examples, the configuration data **108** may represent at least a first ranking associated with a first event, a second ranking associated with a second event, a third ranking associated with a third event, and/or so forth. For example, if the device **102** is used to process data corresponding to ranking frames of image data, the configuration data **108** may indicate or include rankings associated with slow motion being detected, fast motion being detected, an object being detected, low light being detected, custom weightage, no motion being detected, and/or any other ranking types associated with other event types, and/or the like. In some examples, the device **102** may be preconfigured with the configuration data **108**. Additionally, or alternatively, in some examples, one or more users may be able to generate and/or update the configuration data **108** to include additional and/or alternative events for ranking sensor data **104**. For example, the user(s) may be able to customize the configuration data **108** based on processing tasks that may be used when processing on the sensor data **104**.

[0030] For instance, FIG. **2** illustrates an example of configuration data **202** (which may represent, and/or be similar to, the configuration data **108**) that may be used for ranking sensor data (e.g., sensor data **104**), in accordance with some embodiments of the present disclosure. As shown, the configuration data **202** may associate various events **204**(**1**)-(N) (also referred to singularly as "event **204**" or in plural as "events **204**") with different rankings **206**(**1**)-(N) (also referred to singularly as "ranking **206**" or in plural as "rankings **206**"). For a first example, if the configuration data **202** is associated with image data, then the events **204** may include an object is detected, a type of object that is detected, motion is detected, slow motion is detected, fast motion is detected, a close object is detected, a distant object is detected, low light is detected, and/or any other events associated with image detection. For a second example, if the configuration data **202** is associated with audio data, then the events **204** may include sound is detected, speech is detected, specific speech (e.g., specific words) is detected, and/or any other event associated with audio data.

[0031] The rankings **206** may include numerical rankings, alphabetical rankings, alphanumeric rankings, characters, syllables, symbols, and/or any other type ranking. For a first example, the first ranking **206**(**1**) may include a first number (e.g., 1), the second ranking **206**(**2**) may include a second number (e.g., 2), the third ranking **206**(**3**) may include a third number (e.g., 3), and/or so forth. For a second example, the first ranking **206**(**1**) may include a first letter (e.g., A), the second ranking **206**(**2**) may include a second letter (e.g., B), the third ranking **206**(**3**) may include a third letter (e.g., C), and/or so forth. While these are just a few examples of rankings that may be used for sensor data, in other examples, any other type of ranking may be used for the sensor data.

[0032] While the example of FIG. **2** illustrates that each event **204** is associated with a single ranking **206**, in other examples, one or more of the events **204** may be associated with multiple rankings. For example, the first event **204**(**1**) may be associated with the first ranking **206**(**1**) along with at least one other ranking. Additionally, in some examples, more than one event **204** may associated with the same ranking **206**. For example, the first ranking **206**(**1**) associated with the first event **204**(**1**) may be the same as the second ranking **206**(**2**) associated with the second event **204**(**2**).

[0033] Referring back to the example of FIG. **1**, and with regard to the ranking, the process **100** may include the device **102** using one or more detection components **110** processing at least a portion of the sensor data **104** in order to detect events. As described herein, in some examples, the detection component(s) **110** may process the same sensor data **104** for which the device **102** is processing or determining a ranking when detecting the events. In some examples, when processing the sensor data **104** that is being ranked, the detection component(s) **110** may use one or more machine learning models, one or more neural networks, one or more algorithms, one or more modules, and/or any other type of processing component. Additionally, as will be described in more detail herein, the processing component(s) used by the detection component(s) **110** may require less computing resources as compared to one or more processing components later used by one or more systems **112** that then use the rankings to process the sensor data **104**.

[0034] For a first example of processing sensor data **104**, and if the sensor data **104** includes image data, the detection component(s) **110** may process at least a portion of the image data to determine whether a frame is associated with one or more events. For instance, the detection component(s) **110** may determine whether the frame depicts an object, depicts a type of object, does not depict an object, depicts low light, is associated with motion, is associated with a specific pace of motion (e.g., slow motion, fast motion, etc.), and/or so forth. For a second example of processing sensor data **104**, and if the sensor data **104** includes audio data, the detection component(s) **110** may process at least a portion of the audio data to determine whether an audio frame is associated with one or more events. For instance, the detection component(s) **110** may determine whether the audio frame is associated with sound, is not associated with sound, associated with speech, is associated with one or more words of speech, is associated with speech from a specific user, is not associated with speech, and/or so forth.

[0035] For instance, FIG. **3** illustrates an example of processing image data to detect events, in accordance with some embodiments of the present disclosure. In the example of FIG. **3**, the detection component(s) **110** may be processing image data representing at least frames **302**(**1**)-(**6**) (also referred to singularly as “frame **302**” or in plural as “frames **302**”). As shown, the frames **302** may depict at least an environment **304**, where various objects may move within the environment **302**. As such, based at least on the processing, the detection component(s) **110** may determine that the first frame **302**(**1**) is associated with a detected object **306**(**1**) and/or motion, the second frame **302**(**2**) is associated with the detected object **306**(**1**) and/or motion, the third frame **302**(**3**) is associated with the detected object **306**(**1**) and/or motion, the fourth frame **302**(**4**) is associated with no object detection and/or no motion, the fifth frame **302**(**5**) is associated with no object detection and/or no motion, and the sixth frame **302**(**6**) is associated with detected objects **306**(**2**)-(**3**) and/or motion.

[0036] In some examples, the detection component(s) **110** may determine additional information associated with one or more of the frames **302** based at least on the processing. For example, the detection component(s) **110** may determine a first type (e.g., person) associated with the object **306**(**1**), a second type (e.g., vehicle) associated with the objects **306**(**2**)-(**3**), bounding shapes (e.g., bounding boxes, bounding circles, bounding hexagons, etc.) associated with the objects **306**(**1**)-(**3**) as depicted in the images **302**, directions of travel associated with the objects **306**(**1**)-(**3**), velocities associated with the objects **306**(**1**)-(**3**), amounts of light associated with the frames **302** and/or the environment **304**, and/or any other information.

[0037] Referring back to the example of FIG. **1**, in addition to, or alternatively from, the detection component(s) **110** detecting events using the sensor data **104** that is being ranked, in some examples, the detection component(s) **110** may detect events using additional sensor data **104** for which the device **102** may not be ranking. For instance, the detection component(s) **110** may process the additional sensor data **104** in order to detect objects, detect motion of objects, detect no objects, detect sound, detect speech, detect an amount of light, detect a temperature, and/or detect any other type of event. For a first example, the detection component(s) **110** may process motion data, LiDAR data, and/or RADAR data in order to detect objects and/or motion of objects. For a second example, the detection component(s) **110** may process audio data in order to detect sound, speech, and/or any other noise events. While these are just a few examples of the detection component(s) **110** using additional sensor data **104** to detect events, in other examples, the detection component(s) **110** may detect any other type of event using the additional sensor data **104**.

[0038] In some examples, the detection component(s) **110** may be configured to process an entirety of the sensor data **104** to detect events. For example, if the sensor data **104** includes image data, the detection component(s) **110** may be configured to process all of the frames of the image data in order to detect respective events associated with each of the frames. In some examples, the detection component(s) **110** may be configured to process only a portion of the sensor data **104** to detect events. For example, and again if the sensor data **104** includes image data, the detection component(s) **110** may be configured to process every other frame, or frames at any other interval or periodicity, and/or the like to detect respective events associated with the processed frames. In such examples, the detection component(s) **110** may only process a portion of the sensor data **104** in order to conserve computing resources.

[0039] The process **100** may then include the device **102** using a ranking component **114** to rank at least a portion of the sensor data **104** based at least on the events detected by the detection component(s) **110**. As described herein, in some examples, the ranking component **114** may be configured to determine a respective ranking for each sensor representation (e.g., each frame) of the sensor data **104**. However, in other examples, the ranking component **114** may be configured to determine a respective ranking for groups of sensor representations (e.g., groups of frames) of the sensor data **104**. For example, if the ranking component **114** determines a ranking associated with a sensor representation associated with the group of sensor representations, then the ranking component **114** may associate with entire group of sensor representations with the ranking. In such an example, the ranking component **114** may determine the ranking for the first sensor representation, an intermediate sensor representation, the last sensor representation, and/or any other sensor representation included in the group of sensor representations.

[0040] To determine a ranking for a sensor representation (and/or a group of sensor representations), the ranking component **114** may use the output from the detection component(s) **110** to determine an event associated with the sensor representation. The ranking component **114** may then use the event, along with the configuration data **108**, to determine the ranking for the sensor representation. For a first example, if a first event corresponding to a first sensor representation is associated with a first ranking, then the ranking component **114** may associate the first sensor representation with the first ranking. Additionally, if a second event corresponding to a second sensor representation is associated with a second ranking, then the ranking component **114** may associate the second sensor representation with the second ranking. For a second example, if a first event corresponding to a first group of sensor representations is associated with a first ranking, then the ranking component **114** may associate the sensor representations included in the first group of sensor representations with the first ranking. Additionally, if a second event corresponding to a second group of sensor representations is associated with a second ranking, then the ranking component **114** may associate the sensor representations included in the second group of sensor representations with the second ranking.

[0041] For instance, FIG. **4** illustrates an example of ranking the frames **302** based at least on detected events, in accordance with some embodiments of the present disclosure. As shown, the ranking component **114** may associate first image data **402**(**1**) representing the first frame **302**(**1**) with a first ranking **404**(**1**), second image data **402**(**2**) representing the second frame **302**(**2**) with a second ranking **404**(**2**), third image data **402**(**3**) representing the third frame **302**(**3**) with a third ranking **404**(**3**), fourth image data **402**(**4**) representing the fourth frame **302**(**4**) with a fourth ranking **404**(**4**), fifth image data **402**(**5**) representing the fifth frame **302**(**5**) with a fifth ranking **404**(**5**), and sixth image data **402**(**6**) representing the sixth frame **302**(**6**) with a sixth ranking **404**(**6**). In some examples, one or more of the ranking **404**(**1**)-(**6**) (which may also be referred to singularly as "ranking **404**" or in plural as "rankings **404**") may be similar to one another.

[0042] For a first example, the rankings **404**(**1**)-(**3**) and **404**(**6**) may be similar to one another based at least on the events associated with the frames **302**(**1**)-(**3**) and **302**(**6**) indicating a detected object **306**(**1**)-(**3**) and/or motion of a detected object **306**(**1**)-(**3**). For a second example, the rankings **404**(**1**)-(**3**) may be similar to one another based at least on the events associated with the frames **302**(**1**)-(**3**) indicating a same type of detected object **306**(**1**). However, the sixth ranking **404**(**6**) may be different than the rankings **404**(**1**)-(**3**) based at least on the event associated with the sixth frame **302**(**6**) indicating a second type of detected objects **306**(**2**)-(**3**). For a third example, the rankings **404**(**4**)-(**5**) may be similar to one another based at least on the events associated with the frames **302**(**4**)-(**5**) indicating that no objects and/or motion were detected. While these are just a few examples of rankings **404** that may be similar to one another, in other examples, other rankings **404** may be similar to one another based at least on the events associated with the frames **302**.

[0043] Referring back to the example of FIG. **1**, in some examples, the ranking component **114** may perform additional processing when ranking the sensor data **104**. For instance, if the ranking component **114** is ranking first sensor data **104** generated using a first sensor **106** based at least on events detected using second sensor data **104** generated using a second sensor **106**, then the ranking component **114** may initially determine which events detected using the second sensor data **104** to associate with portions of the first sensor data **104**. For example, and for an event, the ranking component **114** may use second timestamps associated with the second sensor data **104** to determine a time that the event was detected. The ranking component **114** may then use first timestamps associated with the first sensor data **104** to determine a portion of the first sensor data **104** that is associated with the time. For instance, the ranking component **114** may determine that the portion of the first sensor data **104** was generated at the time and/or approximate to the time.

The ranking component **114** may then associate that portion of the first sensor data **104** with the event.

[0044] In other words, the ranking component **114** may use the first timestamps and/or the second timestamps to align the first sensor data **104** with the second sensor data **104**. The ranking component **114** may then use the alignment to associate the events detected using the second sensor data **104** with the first sensor data **104**. In some examples, an event detected using the second sensor data **104** may be associated with a single sensor representation of the first sensor data **104**. In some examples, an event detected using the second sensor data **104** may be associated with multiple sensor representations of the first sensor data **104**.

[0045] As further shown by the example of FIG. **1**, the process **100** may include the device **102** using the ranking component **114** and/or an encoding component **118** to associate the rankings, which are represented by ranking data **116**, with the sensor data **104**. In some examples, to associate a ranking with a portion of the sensor data **104**, the ranking component **114** and/or the encoding component **118** may embed the ranking as a message (e.g., a SEI message) and/or as part of a dedicated field associated with the portion of the sensor data **104** in a bitstream. For example, if the sensor data **104** includes image data, then the ranking component **114** and/or the encoding component **118** may embed the ranking as a message and/or as part of a dedicated field associated with a frame of the image data in a bitstream. In some examples, such as when the ranking component **114** and/or the encoding component **118** is embedding the ranking to a group of sensor representations, then the ranking component **114** and/or the encoding component **118** may perform similar processes to embed the ranking into one or more (e.g., each) of the sensor representations included in the group of sensor representations.

[0046] The process **100** may also include the encoding component **118** encoding the sensor data **104** in order to generate encoded sensor data **120**. In some examples, the encoding component **118** may encode the sensor data **104** after embedding the ranking while, in other examples, the encoding component **118** may encode the sensor data **104** before embedding the ranking. In either example, after performing the encoding, the encoded sensor data **120** may represent the sensor data **104** encoded with the embedded ranking. While the example of FIG. **1** describes embedding the ranking into the sensor data **104** and/or the encoded sensor data **120**, in other examples, the device **102** may keep the ranking data **116** separate from the sensor data **104** and/or the encoded sensor data **104**.

[0047] The process **100** may then include the device(s) **102** sending the encoded sensor data **120**, the ranking data **116**, and/or the sensor data **104** to the system(s) **112** for processing. In some examples, such as when the system(s) **112** processes data from multiple devices **102**, one or more (e.g., each) of the devices **102** may send the encoded sensor data **120**, the ranking data **116**, and/or the sensor data **104** to the system(s) **112**. As described herein, the system(s) **112** may be configured to process the sensor data **104** using one or more processing tasks, such as processing tasks that are associated with a processing pipeline. For a first example, if the system(s) **112** is processing image data using an image processing pipeline, then the system(s) **112** may process the image data to perform one or more inferencing tasks, one or more object detection tasks, one or more object tracking tasks, one or more lighting estimation or simulation tasks, and/or any other type of processing task that may be performed on image data. For a second example, if the system(s) **112** is processing audio data using an audio processing pipeline, then the system(s) **112** may process the audio data to perform one or more speech recognition tasks, one or more voice recognition tasks, one or more natural language understanding tasks, one or more speaker recognition tasks, and/or any other type of processing task that may be performed on audio data.

[0048] As shown, the process **100** may include the system(s) **112** using one or more decoder components **122** to decode the encoded sensor data **120** in order to again generate (e.g., obtain, retrieve, etc.) the sensor data **104**. For instance, the system(s) **112** may use a respective decoder component **122** to decode the encoded sensor data **120** from one or more (e.g., each) of the device(s) **102**. The process **100** may then include the system(s) **112** using a batching component **124** to perform batch processing associated with the sensor data **104**. For instance, the batching component **124** may perform batching by at least collecting and/or combining the sensor data **104** from the different devices **102**. In some examples, the batching component **124** may further perform the batch processing by aligning the sensor data **104** from the devices **102** with one another, such as based on timestamps associated with the sensor data **104**. By performing batch processing according to such processes, the system(s) **112** may be able to process the sensor data **104** for the devices **102** together using the processing task(s) (e.g., using the processing pipeline).

[0049] The process **100** may include the system(s) **112** using a determination component **126** that is configured to determine how to process the sensor data **104**. As described herein, in some examples, the determination component **126** may use the rankings associated with the sensor data **104** to determine how to process portions of the sensor data **104** using one or more processing components **128**, where the processing component(s) **128** is associated with performing the one or more processing tasks. For example, and for a portion of the sensor data **104** (e.g., one or more frames of the sensor data **104**), the determination component **126** may determine to process the portion of the sensor data **104** using one or more of the processing component(s) **128** based at least on the ranking associated with the portion of the sensor data **104** and/or determine to refrain from processing the portion of the sensor data **104** using one or more of the processing component(s) **128** based at least on the ranking.

[0050] In some examples, the determination component **126** may additionally using configuration data **130** to determine how to process the portions of the sensor data **104** using the processing component(s) **128**. For instance, the configuration data **130** may associate respective rankings with one or more processing component(s) **128** for processing the sensor data **104**. For example, the configuration data **130** may associate a first ranking with one or more first processing components **128**, a second ranking with one or more second processing components **128**, a third ranking with one or more third processing components **128**, and/or so forth.

[0051] For a first example, if the sensor data **104** includes image data, then the determination component **126** may determine to process one or more first frames of the image data to perform one or more inferencing operations, one or more object detection operations, and/or one or more object tracking operations based at least on the first frame(s) being associated with one or more first rankings. For instance, the first ranking(s) may be associated with one or more first events such as an object being detected, motion being detected, and/or any other event that is associated with the first frame(s) depicting an object. In some examples, the configuration data **130** may associate the first ranking(s) with inferencing, object detection, and/or object tracking. Additionally, the determination component **126** may determine to refrain from processing one or more second frames of the image data—and refrain from performing one or more of the inferencing operations, one or more of the object detection operations, and/or one or more of the object tracking operations based at least on the second frame(s) being associated with one or more second rankings. For instance, the second ranking(s) may be associated with one or more second events that indicate no object and/or no motion was detected. In some examples, the configuration data **130** may associate the second ranking(s) with no processing tasks and/or processing tasks other than inferencing, object detection, and/or object tracking.

[0052] For a second example, and again if the sensor data **104** includes image data, then the determination component **126** may determine to process one or more first frames of the image data using a first tracking component based at least on the first frame(s) being associated with one or more first rankings. For instance, the first ranking(s) may be associated with one or more first events such as detection of only a single object. In some examples, the configuration data **130** may associate the first ranking(s) with the first tracking component. Additionally, the determination component **126** may determine to process one or more second frames of the image data using a second tracking component based at least on the second frame(s) being associated with one or more second rankings. For instance, the second ranking(s) may be associated with one or more second events such as a detection of multiple objects. In some examples, the configuration data **130** may associate the second ranking(s) with the second tracking component. Additionally, in such an example, the second tracking component may use higher quality object trackers as compared to the first tracking component such that the second tracking component is better at tracking multiple objects. However, the second tracking component may also use a greater amount of computing resources as compared to the first tracking component.

[0053] For a third example, if the sensor data **104** includes audio data, then the determination component **126** may determine to process one or more first audio frames of the audio data to perform sound recognition and/or speech recognition based at least on the first audio frame(s) being associated with one or more first rankings. For instance, the first ranking(s) may be associated with one or more first events such as the first audio frame(s) being associated with detected sound. In some examples, the configuration data **130** may associate the first ranking(s) with the sound recognition and/or the speech recognition. Additionally, the determination component **126** may determine to refrain from processing one or more second audio frames of the audio data to perform one or more of the sound recognition operations and/or one or more of the speech recognition operations based at least on the second audio frame(s) being associated with one or more second rankings. For instance, the second ranking(s) may be associated with one or more second events that indicate no sound was detected. In some examples, the configuration data **130** may associate the second ranking(s) with no processing tasks and/or processing tasks other than sound recognition and/or speech recognition.

[0054] The process **100** may then include the system(s) **112** processing portions of the sensor data **104** based at least on the output from the determination component **126**. For instance, and for a portion of the sensor data **104**, the system(s) **112** may process the portion of the sensor data **104** using one or more of the processing component(s) **128** for which the determination component **126** selected for processing. As described herein, a processing component **128** may include one or more machine learning models, one or more neural networks, one or more algorithms, one or more modules, one or more software applications, one or more hardware components, and/or any other type of computing resources that is configured to process sensor data **104** to perform a task.

[0055] For a first example, if a ranking indicates that one or more frames of image data are associated with a detected object, and the determination component **126** then determines that the frame(s) should be processed using inference, object detection, and/or object tracking, then the system(s) **112** may process the frame(s) using a first processing component **128** that performs inferencing, a second processing component **128** that performs object detection, and/or a third processing component **128** that performs object tracking. For a second example, if a ranking indicates that one or more frames of image data are not associated with a detected object, and the determination component **126** then determines that the frame(s) should not be processed using inference, object detection, and/or object tracking, then the system(s) **112** may refrain from processing the frame(s) using a first processing component **128** that performs inferencing, a second processing component **128** that performs object detection, and/or a third processing component **128** that performs object tracking.

[0056] In some examples, the system(s) **112** may refrain from processing a portion of sensor data **104** using a processing component **128** by not applying the portion of the sensor data **104** to the processing component **128**. In some examples, the system(s) **112** may refrain from processing a portion of the sensor data **104** using a processing component **128** by still applying the portion of the sensor data **104** to the processing component **128**, but causing the processing component **128** to refrain from performing any processing (e.g., the portion of the sensor data **104** just passes through the processing component **128**). In such examples, rather than processing the portion of the sensor data **104**, the processing component **128** may generate a set output, such as an output that is similar to a previous portion of the sensor data **104** that was processed by the processing component **128**.

In other words, the processing component **128** may continue to output the same output until the processing component **128** is caused to again process a new portion of the sensor data **104**.

[0057] For instance, FIG. **5** illustrates an example scenario in which the frames **302** are processed using different processing components **502**(**1**)-(**5**) (which may be referred to singularly as "processing component **502**" or in plural as "processing components **502**") based at least on the rankings **404** associated with the frames, in accordance with some embodiments of the present disclosure. In some examples, the processing components **502** (which may represent, and/or be similar to, the processing component(s) **128**) may be part of an image processing pipeline and perform various image processing tasks associated with the image data **402** representing the frames **302**. For example, the processing components **502** may including one or more inferencing components, one or more detection components, one or more segmentation components, one or more tracking components, one or more video composition components, one or more messaging components, and/or any other type of image processing component.

[0058] As shown, based at least on the rankings **404**(**1**)-(**6**), the image data **402**(**1**)-(**3**) and **402**(**6**) may be processed using the processing components **502**(**1**)-(**2**) without processing the image data **402**(**4**)-(**5**) (which is indicated by the dashed lines). In some examples, the image data **402**(**1**)-(**3**) and **402**(**6**) may be processed using the processing components **502**(**1**)-(**2**) since the respective frames **302**(**1**)-(**3**) and **302**(**6**) are associated with the rankings **404**(**1**)-(**3**) and **404**(**6**) that indicate at least that the frames **302**(**1**)-(**3**) and **302**(**6**) depict objects **306**(**1**)-(**3**) and/or motion. For instance, the processing components **502**(**1**)-(**2**) may be associated with performing inferencing and/or object detection. Additionally, based at least on the rankings **404**(**1**)-(**6**), the image data **402**(**1**)-(**3**) may be processed using the processing component **502**(**3**) without processing the image data **402**(**4**)-(**6**) (which is indicated by the dashed lines). In some examples, the image data **402**(**1**)-(**3**) may be processed using the processing component **502**(**3**) since the respective frames **302**(**1**)-(**3**) are associated with the rankings **404**(**1**)-(**3**) that indicate at least that the frames **302**(**1**)-(**3**) depict motion of a single object **306**(**1**) and/or a slow moving object **306**(**1**). For instance, the processing component **502**(**3**) may be associated with performing object tracking.

[0059] Furthermore, based at least on the rankings **404**(**1**)-(**6**), the image data **402**(**6**) may be processed using the processing component **502**(**4**) without processing the image data **402**(**1**)-(**5**) (which is indicated by the dashed lines). In some examples, the image data **402**(**6**) may be processed using the processing component **502**(**4**) since the frame **302**(**6**) is associated with the ranking **404**(**6**) that indicates at least that the frame **302**(**6**) depicts motion of multiple objects **306**(**2**)-(**3**) and/or fast moving objects **306**(**2**)-(**3**). For instance, the processing component **502**(**4**) may also be associated with performing object tracking, however, the processing component **502**(**4**) may include a higher quality tracker as compared to the processing component **502**(**3**). Moreover, based at least on the rankings **404**(**1**)-(**6**), the image data **402**(**1**)-(**6**) may be processed using the processing component **502**(**6**).

[0060] Referring back to the example of FIG. **1**, the process **100** may include generating output data **132** associated with the processing of the sensor data **104**. For example, if the processing component(s) **128** is associated with performing image processing, then the output data **132** may indicate locations of objects (e.g., bounding shapes) within sensor representations of the sensor data **104**, types of the objects, tracking information associated with the objects, and/or any other information associated with the objects. In some examples, the system(s) **112** may then perform one or more operations with respect to the output data **132**, such as storing the output data **132** and/or sending the output data **132** to one or more computing devices. Additionally, in some examples, the system(s) **112** may then perform additional processing with respect to the sensor data **104**, such as again encoding the sensor data **104** using one or more encoders and/or sending the sensor data **104** (e.g., as again encoded) to one or more other computing devices.

[0061] While the example of FIG. **1** illustrates the device(s) **102** as determining the ranking for the sensor data **104** using the configuration data **108**, the detection component(s) **110**, and/or the ranking component **114**, in other examples, the system(s) **112** may determine the ranking using the configuration data **108**, the detection component(s) **110**, and/or the ranking component **114**. For instance, the system(s) **112** may receive the sensor data **104** (and/or the encoded sensor data **120**) from the device(s) **102** and then perform similar processes as the device(s) **102** to rank the sensor data **104**. Additionally, while the examples herein describe the detection component(s) **110** as performing some of the same types of processing as the processing component(s) **128**, in some examples, the detection component(s) **110** may use less computing resources as compared to the processing component(s) **128**. For example, the processing component(s) **128** may include higher quality machine learning models, neural networks, algorithms, modules, software, hardware, and/or the like as compared to the detection component(s) **110**.

[0062] Now referring to FIGS. **6** and **7**, each block of methods **600** and **700**, described herein, comprises a computing process that may be performed using any combination of hardware, firmware, and/or software. For instance, various functions may be carried out by a processor executing instructions stored in memory. The methods **600** and **700** may also be embodied as computer-usable instructions stored on computer storage media. The methods **600** and **700** may be provided by a standalone application, a service or hosted service (standalone or in combination with another hosted service), or a plug-in to another product, to name a few. In addition, the methods **600** and **700** are described, by way of example, with respect to FIG. **1**. However, these methods **600** and **700** may additionally or alternatively be executed by any one system, or any combination of systems, including, but not limited to, those described herein.

[0063] FIG. **6** is a flow diagram showing a method **600** for ranking sensor data, in accordance with some embodiments of the present disclosure. The method **600**, at block B**602**, may include obtaining sensor data generated using one or more sensors of a machine. For instance, the device **102** may obtain the sensor data **104** generated using the sensor(s) **106**. In some examples, the sensor data **104** may include image data representing one or more frames. In some examples, the device **102** may further receive additional sensor data **104** generated using one or more additional sensors **106**, such as LiDAR data, RADAR data, motion data, audio data, and/or any other type of sensor data. In such examples, the additional sensor data **104** may be generated at approximately a same time as the sensor data **104**.

[0064] The method **600**, at block B**604**, may include determining that one or more portions of the sensor data are associated with one or more events. For instance, the device **102** (e.g., the detection component(s) **110**) may determine that the portion(s) of the sensor data **104** is associated with the event(s). In some examples, the device **102** may make the determination based at least on processing the sensor data **104**. For instance, the device **102** may determine that the portion(s) of the sensor data **104** represents an object, motion of an object, an amount of light, sound, speech, and/or any other type of event. Additionally, or alternatively, in some examples, the device **102** may make the determination based at least on processing the additional sensor data **104**. For instance, the device may determine that the additional sensor data **104** represent an object, motion of an object, an amount of light, sound, speech, and/or any other type of event.

[0065] The method **600**, at block B**606**, may include determining, based at least on the one or more events, one or more rankings associated with the one or more portions of the sensor data. For instance, the device **102** (e.g., the ranking component **114**) may determine the ranking(s) for the portion(s) of the sensor data **104** based at least on the event(s). As described herein, in some examples, the ranking component **114** may use the configuration data **108** to determine the ranking(s), where the configuration data **108** may associate various types of events with different rankings. For instance, the device **102** may use the configuration data **108** to determine that the event(s) associated with the portion(s) of the sensor data **104** is further associated with the ranking(s). In some examples, the device **102** may determine more than one ranking associated with the portion(s) of the sensor data **104**.

[0066] The method **600**, at block B**608**, may include associating the one or more rankings with the one or more portions of the sensor data. For instance, the device **102** (e.g., the ranking component **114** and/or the encoding component **118**) may associate the ranking(s) with the portion(s) of the sensor data **104**. In some examples, to perform the association, the device **102** may generate data representative of the ranking(s) and then associate the data with the sensor data **104**. In some examples, to perform the association, the device **102** may embed the ranking(s) into the portion(s) of the sensor data **104**. In either example, a portion of the sensor data **104** may include one or more sensor representations (e.g., one or more frames) of the sensor data **104**.

[0067] The method **600**, at block B**610**, may include sending the sensor data associated with the one or more rankings to one or more computing devices. For instance, the device **102** may send the sensor data **104** that is associated with the ranking(s) to the system(s) **112**. In some examples, before sending the sensor data **104**, the device **102** (e.g., the encoding component **118**) may initially encode the sensor data **104** in order to generate encoded sensor data **120**. The device **102** may then send the encoded sensor data **120** to the system(s) **112**.

[0068] FIG. **7** is a flow diagram depicting a method **700** for using one or more ranking associated with sensor data in order to process the sensor data, in accordance with some embodiments of the present disclosure. The method **700**, at block B**702**, may include receiving sensor data generated using one or more sensors of a device, the sensor data associated with one or more rankings. For instance, the system(s) **112** may receive the sensor data **104** from the device **102**. As described herein, in some examples, the system(s) **112** may receive the sensor data **104** as encoded sensor data **120**. Additionally, in some examples, the ranking(s) may be embedded within the portion(s) of the sensor data **104**. For example, the ranking(s) may be embedded using one or more messages and/or one or more dedicated fields associated with the portion(s) of the sensor data **104**.

[0069] The method **700**, at block B**704**, may include determining, based at least on the one or more rankings, to process one or more portions of the sensor data to perform one or more processing tasks. For instance, the system(s) **112** (e.g., the determination component **126**) may determine one or more of the processing component(s) **128** to use to process the portion(s) of the sensor data **104** based at least on the ranking(s). In some examples, the system(s) **112** may determine the processing component(s) **128** using configuration data **130**. For instance, the configuration data **130** may indicate which of the processing component(s) **128** to use to process the portion(s) of the sensor data **104** based at least on the ranking(s) associated with the portion(s) of the sensor data **104**. In some examples, the system(s) **112** may further determine one or more of the processing component(s) **128** that should not be used to process the portion(s) of the sensor data **104** based at least on the ranking(s).

[0070] The method **700**, at block B**706**, may include causing the one or more portions of the sensor data to be processed using the one or more processing tasks to generate output data and the method **700**, at block B**708**, may include performing one or more operations using the output data. For instance, the system(s) **112** may cause the processing component(s) **128** to process the portion(s) of the sensor data **104** in order to generate the output data **132**. As described herein, the system(s) **112** may process the portion(s) of the sensor data **104** using the processing component(s) **128** as part of a processing pipeline (e.g., in an order). The system(s) **112** may then perform one or more operations using the output data **132**, such as sending the output data **132** to one or more other devices.

Example Content Streaming System

[0071] Now referring to FIG. **8**, FIG. **8** is an example system diagram for a content streaming system **800**, in accordance with some embodiments of the present disclosure. FIG. **8** includes application server(s) **802** (which may include similar components, features, and/or functionality to the example computing device **900** of FIG. **9** and/or which may include the system(s) **112**), client device(s) **804** (which may include similar components, features, and/or functionality to the example computing device **900** of FIG. **9** and/or which may include a device **102**), and network(s) **806** (which may be similar to the network(s) described herein). In some embodiments of the present disclosure, the system **800** may be implemented. The application session may correspond to a game streaming application (e.g., NVIDIA GEFORCE NOW), a remote desktop application, a simulation application (e.g., autonomous or semi-autonomous vehicle simulation), computer aided design (CAD) applications, virtual reality (VR) and/or augmented reality (AR) streaming applications, deep learning applications, and/or other application types.

[0072] In the system **800**, for an application session, the client device(s) **804** may only receive input data in response to inputs to the input device(s), transmit the input data to the application server(s) **802**, receive encoded display data from the application server(s) **802**, and display the display data on the display **824**. As such, the more computationally intense computing and processing is offloaded to the application server(s) **802** (e.g., rendering—in particular ray or path tracing—for graphical output of the application session is executed by the GPU(s) of the game server(s) **802**). In other words, the application session is streamed to the client device(s) **804** from the application server(s) **802**, thereby reducing the requirements of the client device(s) **804** for graphics processing and rendering.

[0073] For example, with respect to an instantiation of an application session, a client device **804** may be displaying a frame of the application session on the display **824** based on receiving the display data from the application server(s) **802**. The client device **804** may receive an input to one of the input device(s) and generate input data in response. The client device **804** may transmit the input data to the application server(s) **802** via the communication interface **820** and over the network(s) **806** (e.g., the Internet), and the application server(s) **802** may receive the input data via the communication interface **818**. The CPU(s) may receive the input data, process the input data, and transmit data to the GPU(s) that causes the GPU(s) to generate a rendering of the application session. For example, the input data may be representative of a movement of a character of the user in a game session of a game application, firing a weapon, reloading, passing a ball, turning a vehicle, etc. The rendering component **812** may render the application session (e.g., representative of the result of the input data) and the render capture component **814** may capture the rendering of the application session as display data (e.g., as image data capturing the rendered frame of the application session). The rendering of the application session may include ray or path-traced lighting and/or shadow effects, computed using one or more parallel processing units-such as GPUs, which may further employ the use of one or more dedicated hardware accelerators or processing cores to perform ray or path-tracing techniques—of the application server(s) **802**. In some embodiments, one or more virtual machines (VMs)—e.g., including one or more virtual components, such as vGPUs, vCPUs, etc.—may be used by the application server(s) **802** to support the application sessions. The encoder **816** may then encode the display data to generate encoded display data and the encoded display data may be transmitted to the client device **804** over the network(s) **806** via the communication interface **818**. The client device **804** may receive the encoded display data via the communication interface **820** and the decoder **822** may decode the encoded display data to generate the display data. The client device **804** may then display the display data via the display **824**.

[0074] The systems and methods described herein may be used for a variety of purposes, by way of example and without limitation, for machine control, machine locomotion, machine driving, synthetic data generation, model training, perception, augmented reality, virtual reality, mixed reality, robotics, security and surveillance, simulation and digital twinning, autonomous or semi-autonomous machine applications, deep learning, environment simulation, data center processing, conversational AI, light transport simulation (e.g., ray-tracing, path tracing, etc.), collaborative content creation for 3D assets, cloud computing and/or any other suitable applications.

[0075] Disclosed embodiments may be comprised in a variety of different systems such as automotive systems (e.g., a control system for an autonomous or semi-autonomous machine, a perception system for an autonomous or semi-autonomous machine), systems implemented using a robot, aerial systems, medial systems, boating systems, smart area monitoring systems, systems for performing deep learning operations, systems for performing simulation operations, systems for performing digital twin operations, systems implemented using an edge device, systems incorporating one or more virtual machines (VMs), systems for performing synthetic data generation operations, systems implemented at least partially in a data center, systems for performing conversational AI operations, systems for performing light transport simulation, systems for performing collaborative content creation for 3D assets, systems implemented at least partially using cloud computing resources, and/or other types of systems.

Example Computing Device

[0076] FIG. **9** is a block diagram of an example computing device(s) **900** (which may include, and/or represent, a device **102** and/or the system(s) **112**) suitable for use in implementing some embodiments of the present disclosure. Computing device **900** may include an interconnect system **902** that directly or indirectly couples the following devices: memory **904**, one or more central processing units (CPUs) **906**, one or more graphics processing units (GPUs) **908**, a communication interface **910**, input/output (I/O) ports **912**, input/output components **914**, a power supply **916**, one or more presentation components **918** (e.g., display(s)), and one or more logic units **920**. In at least one embodiment, the computing device(s) **900** may comprise one or more virtual machines (VMs), and/or any of the components thereof may comprise virtual components (e.g., virtual hardware components). For non-limiting examples, one or more of the GPUs **908** may comprise one or more vGPUs, one or more of the CPUs **906** may comprise one or more vCPUs, and/or one or more of the logic units **920** may comprise one or more virtual logic units. As such, a computing device(s) **900** may include discrete components (e.g., a full GPU dedicated to the computing device **900**), virtual components (e.g., a portion of a GPU dedicated to the computing device **900**), or a combination thereof.

[0077] Although the various blocks of FIG. **9** are shown as connected via the interconnect system **902** with lines, this is not intended to be limiting and is for clarity only. For example, in some embodiments, a presentation component **918**, such as a display device, may be considered an I/O component **914** (e.g., if the display is a touch screen). As another example, the CPUs **906** and/or GPUs **908** may include memory (e.g., the memory **904** may be representative of a storage device in addition to the memory of the GPUs **908**, the CPUs **906**, and/or other components). In other words, the computing device of FIG. **9** is merely illustrative. Distinction is not made between such categories as “workstation,” “server,” “laptop,” “desktop,” “tablet,” “client device,” “mobile device,” “hand-held device,” “game console,” “electronic control unit (ECU),” “virtual reality system,” and/or other device or system types, as all are contemplated within the scope of the computing device of FIG. **9**.

[0078] The interconnect system **902** may represent one or more links or busses, such as an address bus, a data bus, a control bus, or a combination thereof. The interconnect system **902** may include one or more bus or link types, such as an industry standard architecture (ISA) bus, an extended industry standard architecture (EISA) bus, a video electronics standards association (VESA) bus, a peripheral component interconnect (PCI) bus, a peripheral component interconnect express (PCIe) bus, and/or another type of bus or link. In some embodiments, there are direct connections between components. As an example, the CPU **906** may be directly connected to the memory **904**. Further, the CPU **906** may be directly connected to the GPU **908**. Where there is direct, or point-to-point connection between components, the interconnect system **902** may include a PCIe link to carry out the connection. In these examples, a PCI bus need not be included in the computing device **900**.

[0079] The memory **904** may include any of a variety of computer-readable media. The computer-readable media may be any available media that may be accessed by the computing device **900**. The computer-readable media may include both volatile and nonvolatile media, and removable and non-removable media. By way of example, and not limitation, the computer-readable media may comprise computer-storage media and communication media.

[0080] The computer-storage media may include both volatile and nonvolatile media and/or removable and non-removable media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules, and/or other data types. For example, the memory **904** may store computer-readable instructions (e.g., that represent a program(s) and/or a program element(s), such as an operating system. Computer-storage media may include, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which may be used to store the desired information and which may be accessed by computing device **900**. As used herein, computer storage media does not comprise signals per se.

[0081] The computer storage media may embody computer-readable instructions, data structures, program modules, and/or other data types in a modulated data signal such as a carrier wave or other transport mechanism and includes any information delivery media. The term “modulated data signal” may refer to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, the computer storage media may include wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media. Combinations of any of the above should also be included within the scope of computer-readable media.

[0082] The CPU(s) **906** may be configured to execute at least some of the computer-readable instructions to control one or more components of the computing device **900** to perform one or more of the methods and/or processes described herein. The CPU(s) **906** may each include one or more cores (e.g., one, two, four, eight, twenty-eight, seventy-two, etc.) that are capable of handling a multitude of software threads simultaneously. The CPU(s) **906** may include any type of processor, and may include different types of processors depending on the type of computing device **900** implemented (e.g., processors with fewer cores for mobile devices and processors with more cores for servers). For example, depending on the type of computing device **900**, the processor may be an Advanced RISC Machines (ARM) processor implemented using Reduced Instruction Set Computing (RISC) or an x86 processor implemented using Complex Instruction Set Computing (CISC). The computing device **900** may include one or more CPUs **906** in addition to one or more microprocessors or supplementary co-processors, such as math co-processors.

[0083] In addition to or alternatively from the CPU(s) **906**, the GPU(s) **908** may be configured to execute at least some of the computer-readable instructions to control one or more components of the computing device **900** to perform one or more of the methods and/or processes described herein. One or more of the GPU(s) **908** may be an integrated GPU (e.g., with one or more of the CPU(s) **906** and/or one or more of the GPU(s) **908** may be a discrete GPU. In embodiments, one or more of the GPU(s) **908** may be a coprocessor of one or more of the CPU(s) **906**. The GPU(s) **908** may be used by the computing device **900** to render graphics (e.g., 3D graphics) or perform general purpose computations. For example, the GPU(s) **908** may be used for General-Purpose computing on GPUs (GPGPU). The GPU(s) **908** may include hundreds or thousands of cores that are capable of handling hundreds or thousands of software threads simultaneously. The GPU(s) **908** may generate pixel data for output images in response to rendering commands (e.g., rendering commands from the CPU(s) **906** received via a host interface). The GPU(s) **908** may include graphics memory, such as display memory, for storing pixel data or any other suitable data, such as GPGPU data. The display memory may be included as part of the memory **904**. The GPU(s) **908** may include two or more GPUs operating in parallel (e.g., via a link). The link may directly connect the GPUs (e.g., using NVLINK) or may connect the GPUs through a switch (e.g., using NVSwitch). When combined together, each GPU **908** may generate pixel data or GPGPU data for different portions of an output or for different outputs (e.g., a first GPU for a first image and a second GPU for a second image). Each GPU may include its own memory, or may share memory with other GPUs.

[0084] In addition to or alternatively from the CPU(s) **906** and/or the GPU(s) **908**, the logic unit(s) **920** may be configured to execute at least some of the computer-readable instructions to control one or more components of the computing device **900** to perform one or more of the methods and/or processes described herein. In embodiments, the CPU(s) **906**, the GPU(s) **908**, and/or the logic unit(s) **920** may discretely or jointly perform any combination of the methods, processes and/or portions thereof. One or more of the logic units **920** may be part of and/or integrated in one or more of the CPU(s) **906** and/or the GPU(s) **908** and/or one or more of the logic units **920** may be discrete components or otherwise external to the CPU(s) **906** and/or the GPU(s) **908**. In embodiments, one or more of the logic units **920** may be a coprocessor of one or more of the CPU(s) **906** and/or one or more of the GPU(s) **908**.
[0085] Examples of the logic unit(s) **920** include one or more processing cores and/or components thereof, such as Data Processing Units (DPUs), Tensor Cores (TCs), Tensor Processing Units (TPUs), Pixel Visual Cores (PVCs), Vision Processing Units (VPUs), Graphics Processing Clusters (GPCs), Texture Processing Clusters (TPCs), Streaming Multiprocessors (SMs), Tree Traversal Units (TTUs), Artificial Intelligence Accelerators (AIAs), Deep Learning Accelerators (DLAs), Arithmetic-Logic Units (ALUs), Application-Specific Integrated Circuits (ASICs), Floating Point Units (FPUs), input/output (I/O) elements, peripheral component interconnect (PCI) or peripheral component interconnect express (PCIe) elements, and/or the like.
[0086] The communication interface **910** may include one or more receivers, transmitters, and/or transceivers that enable the computing device **900** to communicate with other computing devices via an electronic communication network, included wired and/or wireless communications. The communication interface **910** may include components and functionality to enable communication over any of a number of different networks, such as wireless networks (e.g., Wi-Fi, Z-Wave, Bluetooth, Bluetooth LE, ZigBee, etc.), wired networks (e.g., communicating over Ethernet or InfiniBand), low-power wide-area networks (e.g., LoRaWAN, SigFox, etc.), and/or the Internet. In one or more embodiments, logic unit(s) **920** and/or communication interface **910** may include one or more data processing units (DPUs) to transmit data received over a network and/or through interconnect system **902** directly to (e.g., a memory of) one or more GPU(s) **908**.
[0087] The I/O ports **912** may enable the computing device **900** to be logically coupled to other devices including the I/O components **914**, the presentation component(s) **918**, and/or other components, some of which may be built in to (e.g., integrated in) the computing device **900**. Illustrative I/O components **914** include a microphone, mouse, keyboard, joystick, game pad, game controller, satellite dish, scanner, printer, wireless device, etc. The I/O components **914** may provide a natural user interface (NUI) that processes air gestures, voice, or other physiological inputs generated by a user. In some instances, inputs may be transmitted to an appropriate network element for further processing. An NUI may implement any combination of speech recognition, stylus recognition, facial recognition, biometric recognition, gesture recognition both on screen and adjacent to the screen, air gestures, head and eye tracking, and touch recognition (as described in more detail below) associated with a display of the computing device **900**. The computing device **900** may be include depth cameras, such as stereoscopic camera systems, infrared camera systems, RGB camera systems, touchscreen technology, and combinations of these, for gesture detection and recognition. Additionally, the computing device **900** may include accelerometers or gyroscopes (e.g., as part of an inertia measurement unit (IMU)) that enable detection of motion. In some examples, the output of the accelerometers or gyroscopes may be used by the computing device **900** to render immersive augmented reality or virtual reality.
[0088] The power supply **916** may include a hard-wired power supply, a battery power supply, or a combination thereof. The power supply **916** may provide power to the computing device **900** to enable the components of the computing device **900** to operate.
[0089] The presentation component(s) **918** may include a display (e.g., a monitor, a touch screen, a television screen, a heads-up-display (HUD), other display types, or a combination thereof),

speakers, and/or other presentation components. The presentation component(s) **918** may receive data from other components (e.g., the GPU(s) **908**, the CPU(s) **906**, DPUs, etc.), and output the data (e.g., as an image, video, sound, etc.).

Example Data Center

[0090] FIG. **10** illustrates an example data center **1000** that may be used in at least one embodiments of the present disclosure. The data center **1000** may include a data center infrastructure layer **1010**, a framework layer **1020**, a software layer **1030**, and/or an application layer **1040**.

[0091] As shown in FIG. **10**, the data center infrastructure layer **1010** may include a resource orchestrator **1012**, grouped computing resources **1014**, and node computing resources ("node C.R.s") **1016**(**1**)-**1016**(N), where "N" represents any whole, positive integer. In at least one embodiment, node C.R.s **1016**(**1**)-**1016**(N) may include, but are not limited to, any number of central processing units (CPUs) or other processors (including DPUs, accelerators, field programmable gate arrays (FPGAs), graphics processors or graphics processing units (GPUs), etc.), memory devices (e.g., dynamic read-only memory), storage devices (e.g., solid state or disk drives), network input/output (NW I/O) devices, network switches, virtual machines (VMs), power modules, and/or cooling modules, etc. In some embodiments, one or more node C.R.s from among node C.R.s **1016**(**1**)-**1016**(N) may correspond to a server having one or more of the above-mentioned computing resources. In addition, in some embodiments, the node C.R.s **1016**(**1**)-**10161**(N) may include one or more virtual components, such as vGPUs, vCPUs, and/or the like, and/or one or more of the node C.R.s **1016**(**1**)-**1016**(N) may correspond to a virtual machine (VM).

[0092] In at least one embodiment, grouped computing resources **1014** may include separate groupings of node C.R.s **1016** housed within one or more racks (not shown), or many racks housed in data centers at various geographical locations (also not shown). Separate groupings of node C.R.s **1016** within grouped computing resources **1014** may include grouped compute, network, memory or storage resources that may be configured or allocated to support one or more workloads. In at least one embodiment, several node C.R.s **1016** including CPUs, GPUs, DPUs, and/or other processors may be grouped within one or more racks to provide compute resources to support one or more workloads. The one or more racks may also include any number of power modules, cooling modules, and/or network switches, in any combination.

[0093] The resource orchestrator **1012** may configure or otherwise control one or more node C.R.s **1016**(**1**)-**1016**(N) and/or grouped computing resources **1014**. In at least one embodiment, resource orchestrator **1012** may include a software design infrastructure (SDI) management entity for the data center **1000**. The resource orchestrator **1012** may include hardware, software, or some combination thereof.

[0094] In at least one embodiment, as shown in FIG. **10**, framework layer **1020** may include a job scheduler **1028**, a configuration manager **1034**, a resource manager **1036**, and/or a distributed file system **1038**. The framework layer **1020** may include a framework to support software **1032** of software layer **1030** and/or one or more application(s) **1042** of application layer **1040**. The software **1032** or application(s) **1042** may respectively include web-based service software or applications, such as those provided by Amazon Web Services, Google Cloud and Microsoft Azure. The framework layer **1020** may be, but is not limited to, a type of free and open-source software web application framework such as Apache Spark™ (hereinafter "Spark") that may utilize distributed file system **1038** for large-scale data processing (e.g., "big data"). In at least one embodiment, job scheduler **1028** may include a Spark driver to facilitate scheduling of workloads supported by various layers of data center **1000**. The configuration manager **1034** may be capable of configuring different layers such as software layer **1030** and framework layer **1020** including Spark and distributed file system **1038** for supporting large-scale data processing. The resource manager **1036** may be capable of managing clustered or grouped computing resources mapped to or allocated for support of distributed file system **1038** and job scheduler **1028**. In at least one embodiment, clustered or grouped computing resources may include grouped computing resource **1014** at data center infrastructure layer **1010**. The resource manager **1036** may coordinate with resource orchestrator **1012** to manage these mapped or allocated computing resources.

[0095] In at least one embodiment, software **1032** included in software layer **1030** may include software used by at least portions of node C.R.s **1016**(**1**)-**1016**(N), grouped computing resources **1014**, and/or distributed file system **1038** of framework layer **1020**. One or more types of software may include, but are not limited to, Internet web page search software, e-mail virus scan software, database software, and streaming video content software.

[0096] In at least one embodiment, application(s) **1042** included in application layer **1040** may include one or more types of applications used by at least portions of node C.R.s **1016**(**1**)-**1016**(N), grouped computing resources **1014**, and/or distributed file system **1038** of framework layer **1020**. One or more types of applications may include, but are not limited to, any number of a genomics application, a cognitive compute, and a machine learning application, including training or inferencing software, machine learning framework software (e.g., PyTorch, TensorFlow, Caffe, etc.), and/or other machine learning applications used in conjunction with one or more embodiments.

[0097] In at least one embodiment, any of configuration manager **1034**, resource manager **1036**, and resource orchestrator **1012** may implement any number and type of self-modifying actions based on any amount and type of data acquired in any technically feasible fashion. Self-modifying actions may relieve a data center operator of data center **1000** from making possibly bad configuration decisions and possibly avoiding underutilized and/or poor performing portions of a data center.

[0098] The data center **1000** may include tools, services, software or other resources to train one or more machine learning models or predict or infer information using one or more machine learning models according to one or more embodiments described herein. For example, a machine learning model(s) may be trained by calculating weight parameters according to a neural network architecture using software and/or computing resources described above with respect to the data center **1000**. In at least one embodiment, trained or deployed machine learning models corresponding to one or more neural networks may be used to infer or predict information using resources described above with respect to the data center **1000** by using weight parameters calculated through one or more training techniques, such as but not limited to those described herein.

[0099] In at least one embodiment, the data center **1000** may use CPUs, application-specific integrated circuits (ASICs), GPUs, FPGAs, and/or other hardware (or virtual compute resources corresponding thereto) to perform training and/or inferencing using above-described resources. Moreover, one or more software and/or hardware resources described above may be configured as a service to allow users to train or performing inferencing of information, such as image recognition, speech recognition, or other artificial intelligence services.

Example Network Environments

[0100] Network environments suitable for use in implementing embodiments of the disclosure may include one or more client devices, servers, network attached storage (NAS), other backend devices, and/or other device types. The client devices, servers, and/or other device types (e.g., each device) may be implemented on one or more instances of the computing device(s) **900** of FIG. **9**—e.g., each device may include similar components, features, and/or functionality of the computing device(s) **900**. In addition, where backend devices (e.g., servers, NAS, etc.) are implemented, the backend devices may be included as part of a data center **1000**, an example of which is described in more detail herein with respect to FIG. **10**.

[0101] Components of a network environment may communicate with each other via a network(s), which may be wired, wireless, or both. The network may include multiple networks, or a network of networks. By way of example, the network may include one or more Wide Area Networks (WANs), one or more Local Area Networks (LANs), one or more public networks such as the Internet and/or a public switched telephone network (PSTN), and/or one or more private networks. Where the network includes a wireless telecommunications network, components such as a base station, a communications tower, or even access points (as well as other components) may provide wireless connectivity.

[0102] Compatible network environments may include one or more peer-to-peer network environments—in which case a server may not be included in a network environment—and one or more client-server network environments—in which case one or more servers may be included in a network environment. In peer-to-peer network environments, functionality described herein with respect to a server(s) may be implemented on any number of client devices.

[0103] In at least one embodiment, a network environment may include one or more cloud-based network environments, a distributed computing environment, a combination thereof, etc. A cloud-based network environment may include a framework layer, a job scheduler, a resource manager, and a distributed file system implemented on one or more of servers, which may include one or more core network servers and/or edge servers. A framework layer may include a framework to support software of a software layer and/or one or more application(s) of an application layer. The software or application(s) may respectively include web-based service software or applications. In embodiments, one or more of the client devices may use the web-based service software or applications (e.g., by accessing the service software and/or applications via one or more application programming interfaces (APIs)). The framework layer may be, but is not limited to, a type of free and open-source software web application framework such as that may use a distributed file system for large-scale data processing (e.g., “big data”).

[0104] A cloud-based network environment may provide cloud computing and/or cloud storage that carries out any combination of computing and/or data storage functions described herein (or one or more portions thereof). Any of these various functions may be distributed over multiple locations from central or core servers (e.g., of one or more data centers that may be distributed across a state, a region, a country, the globe, etc.). If a connection to a user (e.g., a client device) is relatively close to an edge server(s), a core server(s) may designate at least a portion of the functionality to the edge server(s). A cloud-based network environment may be private (e.g., limited to a single organization), may be public (e.g., available to many organizations), and/or a combination thereof (e.g., a hybrid cloud environment).

[0105] The client device(s) may include at least some of the components, features, and functionality of the example computing device(s) **900** described herein with respect to FIG. **9**. By way of example and not limitation, a client device may be embodied as a Personal Computer (PC), a laptop computer, a mobile device, a smartphone, a tablet computer, a smart watch, a wearable computer, a Personal Digital Assistant (PDA), an MP3 player, a virtual reality headset, a Global Positioning System (GPS) or device, a video player, a video camera, a surveillance device or system, a vehicle, a boat, a flying vessel, a virtual machine, a drone, a robot, a handheld communications device, a hospital device, a gaming device or system, an entertainment system, a vehicle computer system, an embedded system controller, a remote control, an appliance, a consumer electronic device, a workstation, an edge device, any combination of these delineated devices, or any other suitable device.

[0106] The disclosure may be described in the general context of computer code or machine-useable instructions, including computer-executable instructions such as program modules, being executed by a computer or other machine, such as a personal data assistant or other handheld device. Generally, program modules including routines, programs, objects, components, data structures, etc., refer to code that perform particular tasks or implement particular abstract data types. The disclosure may be practiced in a variety of system configurations, including hand-held devices, consumer electronics, general-purpose computers, more specialty computing devices, etc.

The disclosure may also be practiced in distributed computing environments where tasks are performed by remote-processing devices that are linked through a communications network.

[0107] As used herein, a recitation of “and/or” with respect to two or more elements should be interpreted to mean only one element, or a combination of elements. For example, “element A, element B, and/or element C” may include only element A, only element B, only element C, element A and element B, element A and element C, element B and element C, or elements A, B, and C. In addition, “at least one of element A or element B” may include at least one of element A, at least one of element B, or at least one of element A and at least one of element B. Further, “at least one of element A and element B” may include at least one of element A, at least one of element B, or at least one of element A and at least one of element B.

[0108] The subject matter of the present disclosure is described with specificity herein to meet statutory requirements. However, the description itself is not intended to limit the scope of this disclosure. Rather, the inventors have contemplated that the claimed subject matter might also be embodied in other ways, to include different steps or combinations of steps similar to the ones described in this document, in conjunction with other present or future technologies. Moreover, although the terms “step” and/or “block” may be used herein to connote different elements of methods employed, the terms should not be interpreted as implying any particular order among or between various steps herein disclosed unless and except when the order of individual steps is explicitly described.

Example Paragraphs

[0109] A: A device comprising: one or more processors to: obtain image data generated using one or more image sensors, the image data representative of one or more frames; determine that the image data is associated with an occurrence of an event; determine, based at least on the event, a ranking associated with the image data; and send, to one or more computing devices, the image data along with an indication of the ranking.

[0110] B: The device of paragraph A, wherein the determination that the image data is associated with the event comprises: analyzing the image data; and determining, based at least on the analyzing, that the one or more frames depict at least a portion of the occurrence of the event.

[0111] C: The device of paragraph A or paragraph B, further comprising one or more sensors, and wherein the one or more processors are further to: obtain sensor data generated using the one or more sensors; and detect, based at least on the sensor data, the occurrence of the event during a period of time corresponding to at least a subset of the sensor data, wherein the image data is determined to be associated with the event is based at least on a detection of the occurrence of the event during a period of time corresponding to at least a subset of the sensor data.

[0112] D: The device of any one of paragraphs A-C, wherein the one or more processors are further to: obtain information that associates one or more events with one or more rankings, wherein the determination that the ranking is associated with the image data comprises: identifying, based at least on the information, the event from the one or more events; and determining, based at least on the information, that the event is associated with the ranking from the one or more rankings.

[0113] E: The device of any one of paragraphs A-D, wherein the event comprises one or more of: a depiction of at least a portion of an object in the one or more frames; a depiction of at least a portion of an object corresponding to an object type in the one or more frames; the one or more frames being associated with motion of the object; the one or more frames being associated with a velocity of the motion; a depiction of at least a portion of a number of objects above a threshold number in the one or more frames; a depiction of an amount of light above a threshold amount in the one or more frames; or the one or more frames being associated with sound detected using audio data.

[0114] F: The device of any one of paragraphs A-E, wherein the one or more processors are further to: obtain second image data generated using the one or more image sensors, the second image data representative of one or more second frames; determine that the second image data is associated with a second event; determine, based at least on the second event, a second ranking associated with the second image data that is different than the ranking associated with the image data; and send, to the one or more computing devices, second encoded data corresponding to the second image data and the second ranking.

[0115] G: The device of paragraph F, wherein: the image data represents a first plurality of frames that includes the one or more frames; the ranking is associated with the first plurality of frames; the second image data represents a second plurality of frames that includes the one or more second frames; and the second ranking is associated with the second plurality of frames.

[0116] H: The device of any one of paragraphs A-G, wherein the one or more processors are further to: process the image data using one or more encoders to generate encoded image data; and embed the ranking associated with the image data into at least one of a message associated with the encoded image data or a field associated with a bitstream of the encoded image data, wherein the encoded image data and the embedded ranking is sent to the one or more computing devices.

[0117] I: The device of any one of paragraphs A-H, wherein the ranking causes the one or more computing device to at least one of: perform one or more first image processing tasks using the image data; or refrain from performing one or more second image processing tasks using the image data.

[0118] J: The device of any one of paragraphs A-I, wherein the device is comprised in at least one of: a control system for an autonomous or semi-autonomous machine; a perception system for an autonomous or semi-autonomous machine; a system for performing one or more simulation operations; a system for performing one or more digital twin operations; a system for performing light transport simulation; a system for performing collaborative content creation for 3D assets; a system for performing one or more deep learning operations; a system implemented using an edge device; a system implemented using a robot; a system for performing one or more generative AI operations; a system for performing operations using one or more large language models (LLMs); a system for performing one or more conversational AI operations; a system for generating synthetic data; a system for presenting at least one of virtual reality content, augmented reality content, or mixed reality content; a system incorporating one or more virtual machines (VMs); a system implemented at least partially in a data center; or a system implemented at least partially using cloud computing resources.

[0119] K: A method comprising: obtaining sensor data generated using one or more sensors; determining that the sensor data is associated with an occurrence of an event; determining, based at least on the event, a ranking associated with the sensor data; and sending, to one or more computing devices, the sensor data and an indication of the ranking.

[0120] L: The method of paragraph K, wherein the determining that the sensor data is associated with the event comprises: analyzing the sensor data; and determining, based at least on the analyzing, that the sensor data corresponds to the occurrence of the event.

[0121] M: The method of paragraph K or paragraph L, further comprising: obtaining second sensor data generated using the one or more second sensors; and detecting the occurrence of the event based at least on the second sensor data, wherein the determining that the sensor data is associated with the event is based at least on detecting the occurrence of the event based at least on the second sensor data.

[0122] N: The method of any one of paragraphs K-M, further comprising: obtaining information that associates one or more events with one or more rankings, wherein the determining the ranking is associated with the sensor data comprises: identifying, based at least on the information, the event from the one or more events; and determining, based at least on the information, that the event is associated with the ranking from the one or more rankings.

[0123] O: The method of any one of paragraphs K-N, further comprising: obtaining second sensor data generated using the one or more sensors, the second sensor data representative of one or more second sensor representations; determining that the second sensor data is associated with a second

event; determining, based at least on the second event, a second ranking associated with the second sensor data that is different than the ranking associated with the sensor data; and sending, to the one or more computing devices, the second sensor data along with a second indication of the second ranking.
[0124] P: The method of paragraph O, wherein: the sensor data represents a first plurality of sensor representations; the ranking is associated with the first plurality of sensor representations; the second sensor data represents a second plurality of sensor representations; and the second ranking is associated with the second plurality of sensor representations.
[0125] Q: The method of any one of paragraphs K-P, further comprising: processing the sensor data using one or more encoders to generate encoded sensor data; and embedding the ranking into at least one of a message associated with the encoded sensor data or a field associated with a bitstream corresponding to the encoded sensor data, wherein the sending the sensor data and the data representative of the ranking comprises sending, to the one or more computing devices, the encoded sensor data with the embedded ranking.
[0126] R: The method of any one of paragraphs K-Q, wherein the ranking causes the one or more computing device to at least one of: perform one or more first sensor processing tasks using the sensor data; or refrain from performing one or more second sensor processing tasks using the sensor data.
[0127] S: One or more processors comprising: one or more processing units to send, to one or more computing devices, image data embedded with a ranking associated with one or more frames represented by the image data, wherein the ranking is determined based at least on determining that the one or more frames is associated with an occurrence of an event corresponding to the ranking.
[0128] T: The one or more processors of paragraph S, wherein the one or more processors are comprised in at least one of: a control system for an autonomous or semi-autonomous machine; a perception system for an autonomous or semi-autonomous machine; a system for performing one or more simulation operations; a system for performing one or more digital twin operations; a system for performing light transport simulation; a system for performing collaborative content creation for 3D assets; a system for performing one or more deep learning operations; a system implemented using an edge device; a system implemented using a robot; a system for performing one or more generative AI operations; a system for performing operations using one or more large language models (LLMs); a system for performing one or more conversational AI operations; a system for generating synthetic data; a system for presenting at least one of virtual reality content, augmented reality content, or mixed reality content; a system incorporating one or more virtual machines (VMs); a system implemented at least partially in a data center; or a system implemented at least partially using cloud computing resources.

Claims

1. A device comprising: one or more processors to: obtain image data generated using one or more image sensors, the image data representative of one or more frames; determine that the image data is associated with an occurrence of an event; determine, based at least on the event, a ranking associated with the image data; and send, to one or more computing devices, the image data along with an indication of the ranking.
2. The device of claim 1, wherein the determination that the image data is associated with the event comprises: analyzing the image data; and determining, based at least on the analyzing, that the one or more frames depict at least a portion of the occurrence of the event.
3. The device of claim 1, further comprising one or more sensors, and wherein the one or more processors are further to: obtain sensor data generated using the one or more sensors; and detect, based at least on the sensor data, the occurrence of the event during a period of time corresponding to at least a subset of the sensor data, wherein the image data is determined to be associated with

the event is based at least on a detection of the occurrence of the event during a period of time corresponding to at least a subset of the sensor data.

4. The device of claim 1, wherein the one or more processors are further to: obtain information that associates one or more events with one or more rankings, wherein the determination that the ranking is associated with the image data comprises: identifying, based at least on the information, the event from the one or more events; and determining, based at least on the information, that the event is associated with the ranking from the one or more rankings.

5. The device of claim 1, wherein the event comprises one or more of: a depiction of at least a portion of an object in the one or more frames; a depiction of at least a portion of an object corresponding to an object type in the one or more frames; the one or more frames being associated with motion of the object; the one or more frames being associated with a velocity of the motion; a depiction of at least a portion of a number of objects above a threshold number in the one or more frames; a depiction of an amount of light above a threshold amount in the one or more frames; or the one or more frames being associated with sound detected using audio data.

6. The device of claim 1, wherein the one or more processors are further to: obtain second image data generated using the one or more image sensors, the second image data representative of one or more second frames; determine that the second image data is associated with a second event; determine, based at least on the second event, a second ranking associated with the second image data that is different than the ranking associated with the image data; and send, to the one or more computing devices, second encoded data corresponding to the second image data and the second ranking.

7. The device of claim 6, wherein: the image data represents a first plurality of frames that includes the one or more frames; the ranking is associated with the first plurality of frames; the second image data represents a second plurality of frames that includes the one or more second frames; and the second ranking is associated with the second plurality of frames.

8. The device of claim 1, wherein the one or more processors are further to: process the image data using one or more encoders to generate encoded image data; and embed the ranking associated with the image data into at least one of a message associated with the encoded image data or a field associated with a bitstream of the encoded image data, wherein the encoded image data and the embedded ranking is sent to the one or more computing devices.

9. The device of claim 1, wherein the ranking causes the one or more computing device to at least one of: perform one or more first image processing tasks using the image data; or refrain from performing one or more second image processing tasks using the image data.

10. The device of claim 1, wherein the device is comprised in at least one of: a control system for an autonomous or semi-autonomous machine; a perception system for an autonomous or semi-autonomous machine; a system for performing one or more simulation operations; a system for performing one or more digital twin operations; a system for performing light transport simulation; a system for performing collaborative content creation for 3D assets; a system for performing one or more deep learning operations; a system implemented using an edge device; a system implemented using a robot; a system for performing one or more generative AI operations; a system for performing operations using one or more large language models (LLMs); a system for performing one or more conversational AI operations; a system for generating synthetic data; a system for presenting at least one of virtual reality content, augmented reality content, or mixed reality content; a system incorporating one or more virtual machines (VMs); a system implemented at least partially in a data center; or a system implemented at least partially using cloud computing resources.

11. A method comprising: obtaining sensor data generated using one or more sensors; determining that the sensor data is associated with an occurrence of an event; determining, based at least on the event, a ranking associated with the sensor data; and sending, to one or more computing devices, the sensor data and an indication of the ranking.

12. The method of claim 11, wherein the determining that the sensor data is associated with the event comprises: analyzing the sensor data; and determining, based at least on the analyzing, that the sensor data corresponds to the occurrence of the event.
13. The method of claim 11, further comprising: obtaining second sensor data generated using the one or more second sensors; and detecting the occurrence of the event based at least on the second sensor data, wherein the determining that the sensor data is associated with the event is based at least on detecting the occurrence of the event based at least on the second sensor data.
14. The method of claim 11, further comprising: obtaining information that associates one or more events with one or more rankings, wherein the determining the ranking is associated with the sensor data comprises: identifying, based at least on the information, the event from the one or more events; and determining, based at least on the information, that the event is associated with the ranking from the one or more rankings.
15. The method of claim 11, further comprising: obtaining second sensor data generated using the one or more sensors, the second sensor data representative of one or more second sensor representations; determining that the second sensor data is associated with a second event; determining, based at least on the second event, a second ranking associated with the second sensor data that is different than the ranking associated with the sensor data; and sending, to the one or more computing devices, the second sensor data along with a second indication of the second ranking.
16. The method of claim 15, wherein: the sensor data represents a first plurality of sensor representations; the ranking is associated with the first plurality of sensor representations; the second sensor data represents a second plurality of sensor representations; and the second ranking is associated with the second plurality of sensor representations.
17. The method of claim 11, further comprising: processing the sensor data using one or more encoders to generate encoded sensor data; and embedding the ranking into at least one of a message associated with the encoded sensor data or a field associated with a bitstream corresponding to the encoded sensor data, wherein the sending the sensor data and the data representative of the ranking comprises sending, to the one or more computing devices, the encoded sensor data with the embedded ranking.
18. The method of claim 11, wherein the ranking causes the one or more computing device to at least one of: perform one or more first sensor processing tasks using the sensor data; or refrain from performing one or more second sensor processing tasks using the sensor data.
19. One or more processors comprising: one or more processing units to send, to one or more computing devices, image data embedded with a ranking associated with one or more frames represented by the image data, wherein the ranking is determined based at least on determining that the one or more frames is associated with an occurrence of an event corresponding to the ranking.
20. The one or more processors of claim 19, wherein the one or more processors are comprised in at least one of: a control system for an autonomous or semi-autonomous machine; a perception system for an autonomous or semi-autonomous machine; a system for performing one or more simulation operations; a system for performing one or more digital twin operations; a system for performing light transport simulation; a system for performing collaborative content creation for 3D assets; a system for performing one or more deep learning operations; a system implemented using an edge device; a system implemented using a robot; a system for performing one or more generative AI operations; a system for performing operations using one or more large language models (LLMs); a system for performing one or more conversational AI operations; a system for generating synthetic data; a system for presenting at least one of virtual reality content, augmented reality content, or mixed reality content; a system incorporating one or more virtual machines (VMs); a system implemented at least partially in a data center; or a system implemented at least partially using cloud computing resources.